Chapter 8.

Recall, $\eta_c = \dfrac{h_{3,s} - h_0}{h_3 - h_0}$

$\eta_c' = \dfrac{h_3 - h_{3x}}{h_3 - h_0}$

$\eta_e = \eta_e' \quad (h_{90,s} = h_y)$

Let nozzle to perfectly expanded location. So '10' corresponds to the exit plane.
We can show,

$$\eta_c' = \frac{\psi\left(1 - \left(\eta_c(\psi - 1) + 1\right)^{-1}\right)}{\psi - 1} \qquad \left(\psi = \frac{T_3}{T_0}\right)$$

Typically $\eta_c' > \eta_c$

We saw in chapter 5,6 that we can get more sophisticated in our treatment of η_is
Compression efficiency will be fnct^n of the foll.

- adiabatic efficiency of the oblique shock sys.
- skin friction coeff
- Surface area
- Freestream Mach #
- T_1/T_0 (external compression) $\equiv (\psi_1)$
- $T_3/T_0 \equiv \psi$

© North Carolina State University, All Rights Reserved

Typical real values of $\eta_c \approx 0.4 - 0.7$ $(\eta_c' \approx 0.8 - 0.95)$. Typically η_c in ram/scram is low compared w/ turbo props.

<u>Burner efficiency</u>: Define $\eta_b(x) = \dfrac{T(x) - T_3}{T_{AFT} - T_3}$ (AFT: adiabatic flame temp)

For turbojet where one is just concerned w/ combustion to convert chemical enthalpy to thermal enthalpy, low subsonic combustion is very good

$$\eta_b \approx 0.98$$

For scramjets $\eta_b \approx 0.8 - 0.9$. (typically measured @ '4')

<u>Nozzle</u>: Expansion efficiency is typically pretty good.

$$\eta_c \approx 0.85 - 0.98$$

- Overall efficiency scales as the product of component efficiencies.

$$\eta_o \propto \prod_i \eta_i$$

To get an idea of system performance, choose 3 different options; high, medium & a low component efficiencies.

For high eff: $\eta_c = 0.7$; $\eta_b = 0.9$; $\eta_e = 0.98$

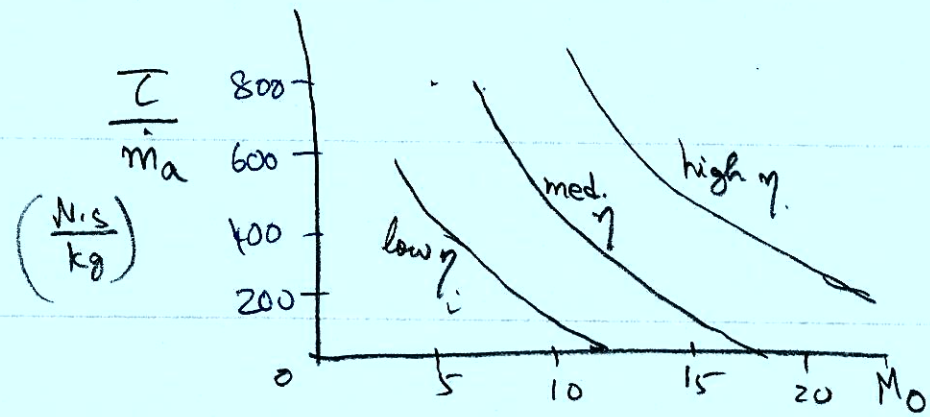
For average eff: $\eta_c = 0.55$; $\eta_b = 0.85$; $\eta_e = 0.91$

For low eff: $\eta_c = 0.4$; $\eta_b = 0.8$; $\eta_e = 0.85$

© North Carolina State University, All Rights Reserved

With this set of η_i s, can look @ overall efficiency w.r.t. M_0

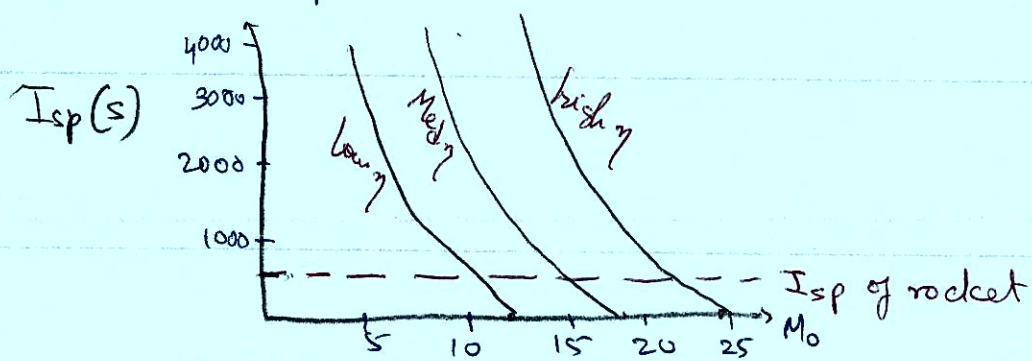
Specific (uninstalled) thrust : $\frac{\tau}{\dot{m}_a}$



⇒ There is a minimum η_i to achieve a given freestream M_0.

- A competing system is the rocket & the rocket thrust is indep. of M_0.
- So @ some Mach #, the rockets will take over & be better performing

To see this, look @ I_{sp}



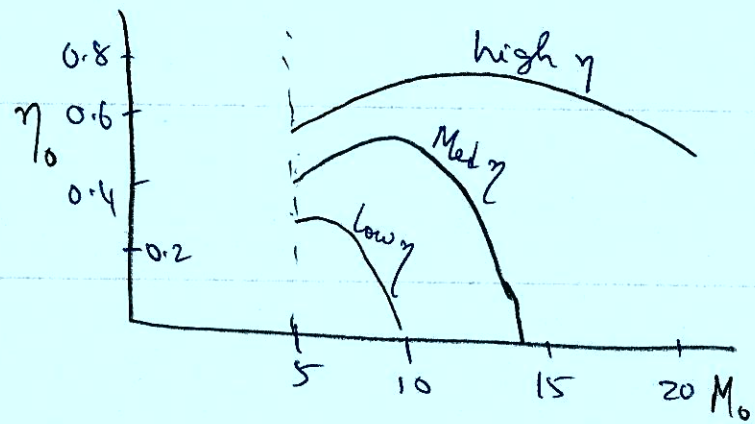
© North Carolina State University, All Rights Reserved

→ The handover from scramjet to rocket occurs @ higher Mach #, as η_i increases

Handover Mach # is a strong non-linear funct of component efficiencies.

- Now, said that $\eta_0 \propto \prod_i \eta_i$

Look @ η_0 vs M_0 for 3 different η_i options



The higher the (η_i)s, the higher the η_0;

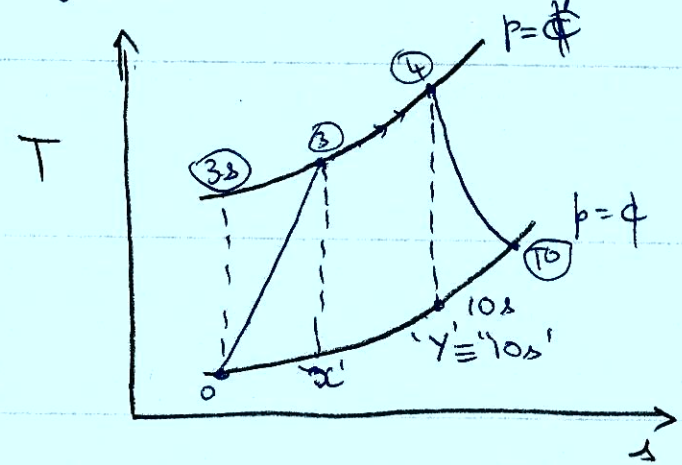
Also the higher the (η_i)s, the η_0 peaks @ a high Mach #.

Another Measure of efficiency can be based on K.E.

© North Carolina State University, All Rights Reserved

η_{KE}, can define component efficiencies based on changes in K.E. across the component.

$$\eta_{KE,c} \equiv \frac{V_x^2}{V_0^2} < 1$$



Invoke the first law of t/d.

$$h_{t4} = h_4 + V_4^2/2$$

$$1 = \frac{h_4}{h_{t4}} + \frac{V_4^2}{2h_{t4}}$$

$$1 = \frac{h_3}{h_{t3}} + \frac{V_3^2}{2h_{t3}}$$

$$\eta_{KE,b} = \frac{V_y^2}{2h_{t4}} \cdot \frac{1}{\frac{V_x^2}{2h_{t3}}} = \frac{V_y^2}{V_x^2} \cdot \left(\frac{h_{t3}}{h_{t4}}\right)$$

If $C_{p_3} = C_{p_4}$ then $\frac{h_{t3}}{h_{t4}} = \frac{T_{t3}}{T_{t4}}$

Now, for this special case, $\eta_{KE,b} = \frac{V_y^2}{V_x^2}\tau^{-1}$; $\tau_b = \frac{T_{t4}}{T_{t3}}$.

$$\eta_{KE,e} \equiv \frac{V_{10}^2}{V_y^2} \quad (\text{'y'} \equiv \text{'10s'})$$

© North Carolina State University, All Rights Reserved

Specific thrust: $\frac{\tau}{\dot{m}_a} = (1+f)\,u_e - u_0$

Total K.E. ratio $= \frac{V_{10}^2}{V_0^2}$

$$\frac{V_{10}^2}{V_0^2} = \frac{V_{10}^2}{V_y^2} \cdot \frac{V_y^2}{V_x^2} \cdot \frac{V_x^2}{V_0^2}$$

$$= \eta_{KE,e} \cdot \eta_{KE,b} \cdot \tau_b \cdot \eta_{KE,c}$$

$$u_e = V_{10} = V_0 \left(\eta_{KE,c} \cdot \eta_{KE,b} \cdot \tau_b \cdot \eta_{KE,e}\right)^{1/2}$$

Define $\eta_{KE,0} = \eta_{KE,c} \cdot \eta_{KE,b} \cdot \eta_{KE,e}$

$$\Rightarrow V_{10} = V_0 \left(\eta_{KE,0} \cdot \tau_b\right)^{1/2}$$

So. $$\boxed{\frac{\tau}{\dot{m}_a} = V_0 \left((1+f)\left(\eta_{KE,0}\,\tau_b\right)^{1/2} - 1 \right)}$$

1st law for combustor.

$$\dot{Q} = \eta_b \dot{m}_f H_{vf} = \dot{m}_4 h_{t4} - \dot{m}_3 h_{t3}$$

$$\dot{m}_4 = \dot{m}_3 + \dot{m}_f = \dot{m}_3 (1+f)$$

© North Carolina State University, All Rights Reserved

$\rightarrow T_{t3} = T_{t0}$

$\rightarrow$ Assume ideal gas w/ componentwise const specific heat.

$\rightarrow h = C_p T \Rightarrow h_3 = C_{p_c} T_3 ; \; h_4 = C_{p_b} T_4 ;$

$\rightarrow h_t = C_p T_t$

We can write the total temp. ratio

$$\boxed{\frac{T_{t4}}{T_{t3}} = \frac{\dot{m}_3 C_{p_c}}{\dot{m}_4 C_{p_b}} + \frac{\dot{m}_f}{\dot{m}_4} \frac{\eta_b H_{vf}}{C_{p_b} T_{t3}}}$$

$$\frac{h_{t4}}{h_{t3}} = \frac{C_{p_b} T_{t4}}{C_{p_c} T_{t3}} = \frac{\dot{m}_3}{\dot{m}_4} + \frac{\dot{m}_f}{\dot{m}_4} \frac{\eta_b H_{vf}}{C_{p_c} T_{t3}}$$

With $\eta_{KE,b} = \frac{V_4^2}{V_x^2} \cdot \frac{h_{t3}}{h_{t4}}$

$$\boxed{\tau_b = (1+f)\left(1 + \frac{\eta_b f H_{vf}}{C_{p_c} T_{t0}}\right) ; \text{ where } \tau_b \equiv \frac{h_{t4}}{h_{t3}}}$$

$$\tau_b = (1+f)\left(\frac{C_{p_c}}{C_{p_b}} + \frac{\eta_b f H_{vf}}{C_{p_b} T_{t0}}\right) ; \text{ where } \tau_b \equiv \frac{T_{t4}}{T_{t3}}$$

© North Carolina State University, All Rights Reserved

$$\boxed{\frac{\mathcal{T}}{\dot{m}_a} = V_0 \left[\left\{ \eta_{KE,0} \left[(1+f)\left(1 + \frac{\eta_b f H_{vf}}{c_p T_{t0}}\right) \right]^{1/2} - 1 \right] \right]}$$

Used $\tau_b \equiv \frac{h_{t4}}{h_{t3}}$

© North Carolina State University, All Rights Reserved